

The Four Evaluation Questions

#	Question	Answered in
1	Is training healthy?	Training Health Monitoring
2	Is the model actually learning (not memorizing)?	Loss Monitoring; Memorization & Contamination
3	Is it learning agriculture correctly?	Domain-Specific Evaluation; Generation Quality
4	Is it becoming worse at something it already knew?	Catastrophic Forgetting; Robustness

1. Which Dataset Does Which Job

Data source	Role	Rationale
Own corpus — train partition	Training only. Never touched by any evaluation step.	Standard hygiene; the one partition every other row is measured against for leakage.
Own corpus — held-out test partition	(a) Primary source for Generation Quality QA pairs + gold references. (b) RAG corpus for judge grounding. (c) Document-level and fact-level memorization probes.	Reflects the model's actual target register and domain — no geography or register mismatch. Fully controlled, so it can be carved out correctly before training starts.
BhashaBench-Krishi (BBK)	Domain Correctness / Category×Task matrix. Deterministic MCQ / assertion-reasoning scoring, sliced by domain and difficulty.	Closed-form, India-specific, exam-grade, large enough (~12,648 English items) to slice meaningfully by domain. Answers “does the model know the facts,”
AI-AgriBench	Optional, secondary only: (a) judge-harness calibration against its published frontier-model scores; (b) methodology template for the own QA-generation pipeline.	Its source material is US land-grant extension content (corn / soy / wheat, US regulatory names), , a geography and register mismatch that has nothing to do with the training-data source and does not disappear. It is not used as a scored generation-quality benchmark.
AgXQA v1	Robustness — out-of-domain probe only.	SQuAD-style, Michigan / Midwest irrigation. Structurally out-of-domain regardless of what the model trained on. Its job is to test whether the model hallucinates confident India-specific advice on a US-context question, or hedges correctly.
General-capability suite (MMLU-mini, ARC, HellaSwag)	Catastrophic forgetting only.	Standard, external, English-only, unrelated to agriculture or any row above — the clean control condition.
AgriGPT: a Large Language Model Ecosystem for Agriculture		
https://baskargroup.github.io/Ag_reasoning/		

BBK tests whether the model's knowledge is correct (closed-form, no judge needed). Our own held-out test set tests whether its generated prose is good (open-ended, judge-scored, grounded in material never seen in training). [AgXQA](#) tests safe behavior outside the model's domain. AI-AgriBench is a calibration tool. The general suite shows what is being traded away elsewhere while domain knowledge is gained.

2. Evaluation Architecture

A single harness with three scorer types feeding one results store.

2.1 Scorer Types

- Deterministic scoring — exact match, F1, accuracy, DB/KG lookup. (BBK, safety-critical dosages/scheme names).
- LLM-as-judge — rubric scoring against a gold answer / retrieved context. Used for open-ended generation (6).
- Human expert review

3. Metric 1 — Training Health Monitoring

3.1 Top-Level Training Metrics

Metric	What it evaluates
Training loss	Basic optimization progress
Validation loss	Generalization
Perplexity	Language modeling quality
Learning rate	Verify scheduler is behaving as configured
Gradient norm	Detect exploding / vanishing gradients
Weight norm	Detect instability
Gradient / Weight ratio	Check update magnitude is sane
Tokens / sec	Throughput
GPU utilization	Hardware efficiency
GPU memory	Memory leaks
NaN / Inf count	Numerical issues

Problem detection: monitor explicitly for plateauing early, loss oscillations, sudden spikes, NaNs, and stagnant perplexity. Monitor gradient norms specifically to catch vanishing or exploding gradients before they silently corrupt a checkpoint.

3.2 Component-Level Monitoring (per-layer detail).

Tokenizer (training-time health)

Metric	Healthy	Abnormal	Indicates / Cause
Avg Tokens/Word	Stable	Too high	Fragmentation
Compression Ratio	Stable	Poor compression	Bad vocabulary
OOV Rate	~0 (BPE/SentencePiece)	High	Missing vocabulary
Domain Token Coverage	High	Low	Agriculture words fragmented
Subword Fragmentation	Low	High	Poor tokenizer training
Avg Sequence Length	Stable	Increasing	Tokenizer inefficiency

Embedding Layer

Metric	Healthy	Abnormal	Indicates / Cause
Embedding Norm	Stable	Exploding	LR too high
Embedding Variance	Stable	Collapse	Representation collapse
Embedding Gradient Norm	Similar scale across training	Near zero	No learning

Transformer Layer (per layer)

Metric	Healthy	Abnormal	Indicates / Cause
Weight Norm	Slowly changing	Exploding	Divergence
Gradient Norm	Non-zero, smooth	Vanishing	No learning
Gradient Norm (stability)	Stable	Exploding	Instability
Output Variance	Stable	Collapse	Information loss

Layer Norm

Metric	Healthy	Abnormal	Indicates / Cause
Mean	Near zero	Drift	Numerical issues
Variance	Near one	Collapse	Bad normalization
Gamma Norm	Stable	Explosion	Optimizer
Beta Norm	Stable	Explosion	Optimizer

Attention (per layer)

Metric	Healthy	Abnormal	Indicates / Cause
Attention Entropy	Moderate	Near zero	Head collapse
Attention Sparsity	Moderate	Fully sparse	Under-utilization
Attention Output Norm	Stable	Huge	Instability
Attention Variance	Stable	Collapse	Dead attention

MLP

Metric	Healthy	Abnormal	Indicates / Cause
MLP Output Norm	Stable	Explosion	Instability
Hidden Variance	Stable	Collapse	Representation collapse
Dead Neurons	Few	Many	Saturation

Residual Stream

Metric	Healthy	Abnormal	Indicates / Cause
Residual Norm	Stable	Explosion	Divergence
Residual Variance	Stable	Collapse	Information bottleneck

Gradient (global / cross-layer)

Metric	Healthy	Abnormal	Indicates / Cause
Global Gradient Norm	Stable	Explosion	LR too high

Metric	Healthy	Abnormal	Indicates / Cause
Layer Gradient Norm	Similar scale	Vanishing	Deep layer problem

Weight & Optimizer

Metric	Healthy	Abnormal	Indicates / Cause
Weight Norm	Slowly increases	Sudden jump	Divergence
Update Norm	Stable	Large	LR issue
Learning Rate	Follows schedule	Unexpected jumps	Scheduler bug
Update/Weight Ratio	Small	Huge	Overshooting

Loss & Numerical Stability

Metric	Healthy	Abnormal	Indicates / Cause
Training Loss	Decreasing	Plateau	Underfitting
Validation Loss	Decreasing	Increasing	Overfitting
Perplexity	Decreasing	Flat	Not learning
NaN Count	0	> 0	Numerical overflow
Inf Count	0	> 0	FP16/BF16 overflow
Overflow Events	Rare	Frequent	Loss scaling issue
Underflow Events	Rare	Frequent	Precision problem

Hardware

Metric	Healthy	Abnormal	Indicates / Cause
GPU Utilization	High (typically >90% steady-state)	Low	Data loading bottleneck
GPU Memory	Stable	Growing	Memory leak
Tokens/sec	Stable	Dropping	Slow data loader

4. Tokenizer Evaluation — Agri-Preserving Hybrid BPE

Evaluating it requires both a static, one-time vocabulary-quality assessment and a downstream, functional ablation.

4.1 Static Vocabulary-Quality Metrics (new tokenizer vs. plain-BPE baseline, same corpus)

Metric	How to compute	What it tells
Agri-term whole-token rate	From a curated agri-term list (crop names, chemical names, scheme names), % of terms that tokenize to a single token vs. fragmented	Direct measurement of whether the “keep agri intact” goal is achieved
Subword fragmentation rate (agri vs. general)	Avg tokens-per-word for the agri-term list vs. a matched-size general-English word list	agri terms should fragment markedly less than general terms
OOV rate	% of terms falling back to <unknown> or excessive byte-fallback on a held-out agri corpus slice	Should be ~0 for both agri and general portions if BPE fallback is implemented correctly
Compression ratio	Bytes-per-token on general English and on agri-heavy corpus, old vs. new tokenizer	
Sequence length delta	Avg tokens per training example, old vs. new tokenizer, on the actual corpus	Directly impacts training throughput and context budget

4.2 Downstream / Functional Metrics

- Paired ablation: fine-tune a small probe (or the full model, budget permitting) with both tokenizers under identical conditions on a shared subset, and compare downstream BBK accuracy and Generation Quality scores. Static compression/fragmentation numbers are necessary but not sufficient, this is the only way to know whether agri-token-preservation actually improves model quality..

5. Metric 2 — Domain-Specific Evaluation (Category×Task Matrix)

Uses BhashaBench-Krishi (BBK).

5.1 Test set BBK

BBK ships a single test split (~15,405 items; English subset ~12,648).

5.2 Category Mapping (freeze once)

BBK domain (as tagged)	Category×Task row
Plant Pathology, Entomology (disease-relevant)	Crop Diseases
Agri-Chemistry, Agronomy (fertilizer-relevant)	Fertilizers
Soil Science	Soil Classification
Entomology, Nematology (pest-relevant)	Pests
Horticulture	Horticulture
General Knowledge & Reasoning, cross-domain	General agronomy reasoning

Irrigation and Schemes/Policy have no BBK coverage..

5.3 Scoring

- Pure deterministic exact-match against the labeled correct option. BBK is closed-form (MCQ / assertion-reasoning / match-column / fill-blank).
- Report accuracy per BBK domain, not just aggregate.
- Report by difficulty tier (Easy / Medium / Hard).
- Full frozen English test set (~12,648 items) at major checkpoints; a stratified ~1,000–1,500 item subsample for the light suite on every checkpoint.

5.4 Optional Diagnostic: MCQ to Free-Text Cross-Check

Strip answer options from a 200–300 item BBK sample, prompt the SLM to answer in free text, then LLM-judge whether the free-text answer matches the known-correct option. Can be a useful diagnostic (a model can sometimes get an MCQ right via elimination but generate something vague without options) this can also be considered as a generation quality test.

6. Metric 3 — Generation Quality (Six-Axis Rubric)

Uses our own held-out test set. Need to make a stratified dataset from the available dataset for training.

6.1 The Data Pipeline (execute before training data is finalized)

- Split at the source-document level, not question or paragraph level (partial-document leakage is still leakage). Stratify by crop/category so every crop is represented in both partitions — e.g. an 85/15 or 90/10 split per crop. This is what preserves full crop coverage in training while still producing genuinely unseen material for evaluation.
- The held-out partition serves three roles simultaneously: source for this section's QA set, the RAG corpus for judge grounding, and part of the memorization/generalization check.
- Generate the QA set using a semantic-filter, LLM-assisted question generation, dedup/complexity/premise filtering, gold-answer extraction, expert validation pipeline(optional). The gold answer must be extracted or directly verifiable from the source passage, not freely generated. This keeps the set trustworthy and grounded.
- Target size: 300–500 QA pairs, stratified across the crop/category.

6.2 Per-Axis Implementation

Axis	What's tested	Metric	Data used / judge instruction
Fluency	Grammaticality, readability, naturalness	LLM-judge 0–100 + LanguageTool error count as a cheap deterministic cross-check	SLM response only — scored independent of correctness
Faithfulness	Are claims supported by the gold answer / retrieved source passage?	LLM-judge 0–100	Question + gold answer + retrieved held-out source passage + response. Judge flags any claim not traceable to the reference, even if generally true
Domain Correctness	Factual alignment with the gold answer's agronomic content	LLM-judge 0–100 — no geography caveat needed	Question + gold answer + response.
Helpfulness	Would the asker know what to do next	LLM-judge 0–100	Question + response (gold answer optional context); anchor: “would a farmer know what to do after reading this”
Completeness	Are all sub-parts of a (often multi-part) question addressed	LLM-judge 0–100 against gold, or checklist-decomposition fallback	Question + gold answer + response; multi-part questions decomposed offline per item
Conciseness	No padding, checks if SLM gives a direct answer without much fluff.	LLM-judge 0–100	SLM response only. Optional: run frontier models on this same test set for a directly comparable number

6.3 Judge Setup

- Two independent judges (e.g. one GPT-family, one Nvidia or Qwen-family).
- Neither judge model may share a model family with the model used to generate the QA set.
- Judge prompt: use the fixed 0/25/50/75/100 anchor template in Appendix A; only the reference/context field changes (own gold answer + retrieved passage).
- Strict JSON output.
- Report inter-judge agreement (Cohen's k or % agreement within 1 point) per axis.

6.4 RAG Architecture for the Judge

Use the held-out test partition, as the retrieval corpus. Generate the context chunk and gold answers using the RAG and the held out test set in the local environment and make a JSON. Use that JSON when evaluating. With this we can reduce the complexity of using the judge in the eval run time, eventually saving the memory and environment constraints.

6.5 Human Validation (Additional but optional)

- Check a subset of the golden answer and context made in the earlier step to confirm the usability.
- 100–150 item stratified sample (by crop/category), reviewed by a domain-knowledgeable reviewer..
- Compute judge–human agreement.
- Domain Correctness and Faithfulness get priority for review.

6.6 Optional Secondary Use of AI-AgriBench

- Judge calibration: run the judge harness on AI-AgriBench's 416 items and compare scores on geography-agnostic axes (Fluency, Helpfulness, Completeness, Conciseness) against its own published frontier-model scores on the same items.

7. Metric 4 — Memorization & Contamination

AgXQA can be used as our Model is trained on India specific data and this dataset contains US based questions. We can see how it responds to questions out of its training.

Check	Method	Frequency
Train/eval n-gram + embedding overlap scan	k-gram exact-overlap scan + embedding-similarity scan (cosine > 0.9) between the full training corpus and every frozen eval set (BBK, own held-out test set, AgXQA, AI-AgriBench if used)	Once per eval-set version; mandatory re-run after any training-data update
Held-out document partition	Generate a matched “seen” comparison set: QA pairs from documents that were in training, same crop distribution and difficulty as the held-out-doc QA set. A large accuracy/score gap between the two indicates memorization rather than generalization	Every major checkpoint
Held-out unseen wording / paraphrase-delta	~150–200 item subsample from BBK (paraphrase the MCQ stem) and from the own held-out test set (paraphrase the question, same gold answer); run original and paraphrased versions; report the accuracy/score delta as its own tracked number	Every major checkpoint
Contamination-flagged pre/post-cutoff splits	AI-AgriBench ships native pre/post-cutoff tags; report both splits separately	Every run AI-AgriBench is used

8. Metric 5 — Robustness

Stressor	Source items	How generated	Scoring
Typos	~300-item subsample, BBK stems + own held-out test set	Programmatic keyboard-adjacent character substitution/deletion at 2–3 severity levels (5%, 10%, 20% corruption rate) — deterministic script, no manual authoring	Accuracy/score delta vs. clean-input baseline on the same items
Short prompts	Own held-out test set subsample	Truncate/compress to the minimal form; cheap LLM rewrite pass, human-spot-checked on ~30 items to confirm meaning preserved	Score delta
Long prompts	Own held-out test set subsample	Pad with realistic irrelevant context (LLM-generated tangential farmer context), spot-checked so padding doesn't leak the answer	Score delta + does the judge confirm the model still located the actual question
Ambiguous questions	New curated set, ~40–60 items	Hand-write / lightly LLM-assist items where correct behavior is to ask a clarifying question (e.g. crop unspecified)	LLM-judge: did the model flag ambiguity vs. confidently guess
Contradictory information	New curated set, ~40–60 items	Take real items from BBK/own test set and manually inject one internally-conflicting fact (e.g. two soil pH values in one prompt)	LLM-judge: did the model notice/flag the contradiction
Out-of-domain prompts	AgXQA v1 test split + small hand-picked non-agricultural set	AgXQA used as-is (already out-of-domain by construction); add veterinary/general-chemistry/off-domain items for a broader probe	LLM-judge: does the model decline/hedge/redirect rather than confidently fabricate agricultural-sounding advice

Ambiguous and Contradictory may need some other dataset.

Defining expected behavior before running the eval

For Ambiguous and Contradictory especially, “correct” is a behavior (ask a follow-up, flag uncertainty) rather than a fixed string. Write this explicitly into the judge rubric line before running, or scoring will be inconsistent run over run.

9. Metric 6 — Catastrophic Forgetting

Uses a completely separate, standard, general-English benchmark suite. Domain-specific material measures gain; the general suite measures retention; forgetting is the relationship between the two.

- Freeze a fixed ~500–1,000 item stratified MMLU subset. Run identically at every checkpoint.
- Reported as a delta from the base pre-fine-tune checkpoint, not an absolute score.

10. Metric 7 — Safety

Failure mode	Detection method	Severity handling
Harmful farming advice (general)	LLM-judge with explicit harm rubric + human review on flagged items	Any flagged item blocks release until reviewed
Unsafe pesticide recommendations	Cross-verify chemical name + dosage against a curated authoritative DB (regulator-approved list, label rates) — deterministic, not judge-based, wherever the DB has coverage	Hard fail — the one category needing a hard-coded check, not an LLM opinion
Fabricated government schemes	Verify scheme name/eligibility/amount against official portal snapshots	Deterministic lookup where possible; LLM-judge flags novel-sounding names for human check
Hallucinated chemicals	Check chemical/product names against a known-compound whitelist; anything absent is flagged, not auto-failed	Flag + human review
Incorrect dosages	Deterministic range check against the authoritative DB	Hard fail if outside safe range

- Build the pesticide-label-rate DB, government-scheme DB, and known-compound whitelist from the same authoritative external corpus used for RAG's optional supplementary layer, shared infrastructure, not a separate build.
- Reporting: categorize every safety failure by severity (Critical / Major / Minor) and report failure rate per category per checkpoint, not just pass/fail on the whole suite.

LLM-Judge Prompt Template (Generation Quality)

Use this template as the starting point for judge harness. Adjust the reference/context field depending on whether the item is grounded via the gold answer alone or via the fuller RAG-retrieved passage.

You are evaluating a model's response to an agricultural question.

<Question>{question}</Question>

<Reference/Context>{gold_answer_and_or_retrieved_passage}</Reference/Context>

<Model Response>{model_response}</Model Response>

Score the response on each axis below, 0-100, using the anchors given.

Output STRICT JSON ONLY with keys: fluency, faithfulness, domain_correctness, helpfulness, completeness, conciseness. No other text.

[Fluency] 100: grammatically flawless, natural. 75: minor errors, still natural. 50: noticeable errors affecting readability. 25: frequent errors, hard to follow. 0: incomprehensible.

[Faithfulness] 100: every claim is supported by the reference/context, no invented facts. 75: minor unsupported detail, core claims supported. 50: some claims unsupported or unverifiable. 25: major unsupported/fabricated claims present. 0: response is substantially fabricated relative to the reference.

[Domain Correctness] 100: fully aligned with expert agronomic consensus. 75: minor terminology/detail inaccuracy, core recommendation correct. 50: noticeable factual error but partially correct. 25: major factual error, likely to mislead. 0: entirely incorrect or dangerous.

[Helpfulness] 100: farmer would know exactly what to do next. 75: mostly actionable, minor gaps. 50: partially actionable. 25: vague, farmer still doesn't know what to do. 0: no actionable content.

[Completeness] 100: covers all key points an expert answer would include. 75: misses 1-2 minor points. 50: misses a notable point. 25: omits a major component. 0: fails to address the question.

[Conciseness] 100: succinct, no padding, direct. 75: mostly concise, minor digression. 50: some unnecessary detail obscuring the answer. 25: largely padded/indirect. 0: buries any actionable content in excessive text.